

Phase 1 — Every Decision Explained

Statistical arbitrage: universe, data, and the absence of leaks

Project *regime-aware stat-arb & honest selection*

S&P 500 Utilities — 2014-2026

Reference document for the oral defence

1. What this document is for

This document revisits **every decision** made during Phase 1 of the project and explains it in detail: what was decided, why, what the alternatives were, and what each choice costs. It is written so you can defend the project at the whiteboard without notes.

Phase 1 produces no signal, no backtest, no P&L. It produces only a **universe of securities and a price series**. That is why it looks innocuous — and exactly why it is dangerous: a mistake made here never raises an error. It produces a backtest that runs, reports a Sharpe ratio, and is wrong. Every later phase silently inherits its defects.

The rule that structures all of Phase 1

At date t , a backtest may use only information genuinely available at date t . Any violation is a **leak** (look-ahead bias), and a leak inflates performance without ever announcing itself.

The difficulty is that the worst leaks do not hide in the signal. They hide in how the data was built — that is, here.

Read in order. Sections 2 and 3 set up vocabulary and the mathematics (skip if you are comfortable with cointegration). Sections 5 to 10 cover the decisions. Section 11 answers the two questions posed at the end of Phase 1. Section 12 is a glossary.

2. The vocabulary, with no shortcuts

2.1 What a pair is

A *pair* is a couple of stocks (A, B) traded **together and in opposite directions**: buy A, short B, or the reverse. You never bet on market direction, only on the **gap** between the two names.

That gap is the **spread**. Its simplest definition:

$$s(t) = \log P_A(t) - \beta \cdot \log P_B(t)$$

The coefficient β is the **hedge ratio**. It answers: for one dollar invested in A, how much of B must be sold to neutralise the shared risk? If $\beta = 1.3$, you short 1.3 dollars of B for every dollar of A.

We work in **log-prices**, not raw prices, for two reasons. First, a difference of log-prices is dimensionless (a log-ratio), hence comparable across pairs trading at very different price levels. Second, returns are additive in logs, which makes the P&L computation direct.

2.2 The strategy in one sentence

If the spread is **stationary** — oscillating around a fixed mean rather than drifting away — then when it moves abnormally far from that mean, we bet it will come back. Sell the spread when high, buy it when low, close on reversion.

The whole strategy therefore rests on a **single assumption**: spread stationarity. Phase 2 tests it, Phase 3 corrects for the fact that we test it hundreds of times, and Phase 4 deals with it ceasing to hold without warning.

2.3 Correlation vs cointegration: the distinction that comes up in interviews

This is **the** foundational question on this project. Two series can be strongly correlated without being cointegrated, and vice versa.

Correlation is about *returns* — short-horizon variation. Saying A and B correlate at 0.9 means their daily moves go the same way. It says **nothing** about the gap between their levels over the long run: two names can rise and fall together every single day while drifting apart indefinitely.

Cointegration is about *levels*. It says there exists a β such that $\log P_A - \beta \log P_B$ is stationary, even though $\log P_A$ and $\log P_B$ individually are not.

Formally: a stock price behaves like a series **integrated of order 1**, written $I(1)$ — its difference (the return) is stationary, but its level drifts without bound. Cointegration is the remarkable case where a linear combination of two $I(1)$ series is $I(0)$, i.e. stationary.

The intuition to give at the whiteboard

Two cointegrated $I(1)$ series share the **same stochastic trend**. β is the exact weight that cancels it by subtraction: what remains is stationary noise around the equilibrium relation.

The classic image: a drunk and their dog. Each walks randomly (each path is $I(1)$, unpredictable). But the leash bounds the distance between them, and that distance is stationary. Correlation describes whether they step at the same instant; cointegration describes the existence of the leash.

Direct consequence for this project: cointegration is not a statistical coincidence, it requires an **economic mechanism** playing the role of the leash. That argument, and only that argument, justifies the decision in section 5.

3. Mathematical foundations

This section states precisely what the rest of the document uses informally: what stationarity is, what cointegration is, why a cointegration *test* is needed rather than an ordinary regression, and why each of the Phase 1 decisions is forced rather than chosen. Proofs are given whenever they are short; the two results whose proofs need functional limit theory are stated with a reference and a sketch.

3.1 Stationarity

Throughout, (Ω, F, P) is a probability space and all processes are real-valued and indexed by t in $\mathbf{Z}$.

Definition 1 (Strict stationarity)

A process (X_t) is *strictly stationary* if for every $k \geq 1$, every $t_1 < \dots < t_k$ and every h in $\mathbf{Z}$, the joint laws agree:

$$(X_{t_1}, \dots, X_{t_k}) \stackrel{d}{=} (X_{t_1+h}, \dots, X_{t_k+h}).$$

Definition 2 (Weak or covariance stationarity)

(X_t) is *covariance stationary* if $E[X_t^2] < \infty$ and there exist μ in $\mathbf{R}$ and $\gamma : \mathbf{Z} \rightarrow \mathbf{R}$ with

$$E[X_t] = \mu \quad \text{and} \quad \text{Cov}(X_t, X_{t+h}) = \gamma(h) \quad \text{for all } t, h.$$

The essential point is that neither the mean nor the autocovariance depends on t : the process has no privileged date.

Strict stationarity with finite second moment implies covariance stationarity; the converse fails in general but holds for Gaussian processes, since a Gaussian law is determined by its first two moments. All tests used here concern **covariance** stationarity.

3.2 Integration order, and why a price is not stationary

Definition 3 (Order of integration)

(X_t) is *integrated of order 0*, written $I(0)$, if it is covariance stationary with a strictly positive spectral density at frequency zero. It is $I(d)$ for an integer $d \geq 1$ if the d -th difference $(1 - L)^d X_t$ is $I(0)$, where L is the lag operator, $LX_t = X_{t-1}$.

Proposition 1 (A random walk is not covariance stationary)

Let $X_0 = 0$ and $X_t = X_{t-1} + \varepsilon_t$ with (ε_t) i.i.d., $E[\varepsilon_t] = 0$, $\text{Var}(\varepsilon_t) = \sigma^2$ in $(0, \infty)$. Then

$$\text{Var}(X_t) = t\sigma^2,$$

which depends on t , so (X_t) is not covariance stationary. It is $I(1)$.

Proof.

Telescoping the recursion gives $X_t = \varepsilon_1 + \dots + \varepsilon_t$. The increments are independent and centred, so variances add:

$$\text{Var}(X_t) = \sum_{s=1}^t \text{Var}(\varepsilon_s) = t\sigma^2.$$

Since $\sigma^2 > 0$ the map $t \rightarrow \text{Var}(X_t)$ is strictly increasing, contradicting Definition 2. Its first difference is $\Delta X_t = \varepsilon_t$, which is i.i.d. and hence $I(0)$. ■

Log-prices of liquid equities are, to a good approximation, $I(1)$: daily log-returns are close to serially uncorrelated with finite variance, while the level wanders without an attracting mean. **This is the whole difficulty.** A generic linear combination of two $I(1)$ series is again $I(1)$, so a spread is generically non-stationary and has no mean to revert to. Pairs trading is only possible in the exceptional case of the next definition.

3.3 Cointegration

Definition 4 (Cointegration, Engle and Granger 1987)

Let $X_t = (X_t^{(1)}, \dots, X_t^{(n)})'$ with every component $I(1)$. The vector process is *cointegrated* if there exists β in $\mathbf{R}^n$, $\beta \neq 0$, such that

$$\beta' X_t \text{ is } I(0).$$

Such a β is a *cointegrating vector*. The set of cointegrating vectors, together with 0, is a linear subspace of $\mathbf{R}^n$; its dimension r is the *cointegration rank*.

Two immediate remarks. First, β is defined only up to scale, since $c\beta$ works for any $c \neq 0$; for a pair one therefore normalises $\beta = (1, -\beta_2)'$, which is exactly the hedge ratio of Section 2. Second, for $n = 2$ the rank is 0 or 1: either the pair is cointegrated with an essentially unique β , or it is not cointegrated at all. Rank only becomes an interesting object for baskets, which is the subject of the Johansen extension.

Theorem 1 (Granger representation theorem, 1987 — stated without proof)

If X_t is $I(1)$ with cointegration rank $r \geq 1$, then it admits a *vector error-correction* representation

$$\Delta X_t = \alpha \beta' X_{t-1} + \sum_{i=1}^p \Gamma_i \Delta X_{t-i} + \varepsilon_t,$$

with α, β of full rank r . Conversely such a representation implies cointegration.

Why Theorem 1 is the economic content of this project

The error-correction term $\alpha \beta' X_{t-1}$ says that **today's deviation from equilibrium predicts tomorrow's change**. That is not a restatement of cointegration — it is the assertion that a forecastable component exists, and α measures how fast the deviation is corrected.

This is the precise sense in which cointegration, unlike correlation, is a *tradable* property. Correlation describes contemporaneous co-movement and implies nothing about future returns; the error-correction representation is a statement about conditional expectations of future increments.

It also locates exactly what Phase 4 is about: a *regime break* is α drifting toward 0, or β itself moving.

3.4 Why a regression is not a test: spurious regression

A natural but wrong instinct is to regress one log-price on the other and read off the t-statistic. The following result explains why that procedure is worthless on I(1) data, and why a dedicated cointegration test is unavoidable.

Theorem 2 (Spurious regression; Granger and Newbold 1974, Phillips 1986)

Let X_t and Y_t be **independent** random walks and consider the OLS regression $Y_t = a + b X_t + u_t$ on T observations. Then, as $T \rightarrow \infty$,

$$\hat{b} \xrightarrow{d} \xi \text{ (a non-degenerate random variable),} \quad R^2 \xrightarrow{d} \eta \in (0, 1),$$

and the usual t-statistic diverges,

$$T^{-1/2} t_{\hat{b}} \xrightarrow{d} \zeta \neq 0, \quad \text{so} \quad |t_{\hat{b}}| \xrightarrow{\mathbb{P}} \infty.$$

Consequently the test of $b = 0$ at any fixed level rejects with probability tending to 1, although the two series are independent by construction.

The mechanism is that under independence the residual u_t is itself I(1). The OLS standard error formula assumes I(0) errors; applied to an I(1) residual it is inconsistent and understates the true dispersion by a factor growing with T . Neither β nor its t-statistic converges to anything usable.

The consequence that shapes the whole pipeline

A high R^2 and a large t-statistic between two log-prices are **evidence of nothing**. The only admissible procedure is to run the regression, then test whether the *residual* is I(0) — which is exactly the Engle-Granger two-step procedure of Phase 2.

It also explains why the critical values must be adjusted: the residual being tested is not observed but estimated, and it was estimated by minimising its own variance. Phase 2 quantifies the resulting size distortion at a factor of 3.5.

3.5 Why dividends force total-return prices

Section 7 argues that raw prices inject a deterministic drift into the spread. Here is the statement that makes the argument binding rather than suggestive.

Proposition 2 (A deterministic trend destroys stationarity)

Let (y_t) be covariance stationary with mean μ , let $\delta \neq 0$, and set $s_t = y_t - \delta t$. Then (s_t) is not covariance stationary.

Proof.

By linearity of the expectation, $E[s_t] = \mu - \delta t$. Since $\delta \neq 0$ this is a non-constant function of t , contradicting Definition 2. ■

Now apply this to prices. Write P^{TR} for the total-return price and P^{raw} for the raw price of a stock paying a continuously compounded dividend yield δ . Reinvesting dividends multiplies wealth by $e^{\delta t}$ relative to holding the raw share, so

$$\log P_t^{\text{raw}} = \log P_t^{\text{TR}} - \delta t.$$

Proposition 3 (The raw-price spread carries a deterministic drift)

Let A and B have dividend yields δ_A, δ_B and suppose the total-return log-prices are cointegrated with hedge ratio β , so that $y_t = \log P_{A,t}^{\text{TR}} - \beta \log P_{B,t}^{\text{TR}}$ is $I(0)$. Then the spread computed on raw prices satisfies

$$s_t^{\text{raw}} = y_t - (\delta_A - \beta \delta_B) t,$$

which is $I(0)$ if and only if $\delta_A = \beta \delta_B$.

Proof.

Substitute the identity above into the definition of the spread:

$$s_t^{\text{raw}} = (\log P_{A,t}^{\text{TR}} - \delta_A t) - \beta (\log P_{B,t}^{\text{TR}} - \delta_B t) = y_t - (\delta_A - \beta \delta_B) t.$$

If $\delta_A \neq \beta \delta_B$, Proposition 2 applies and the raw spread is not stationary; if they are equal the drift term vanishes identically. ■

The condition $\delta_A = \beta \delta_B$ is a knife-edge: it holds on a set of measure zero in the parameter space. On the actual universe, measured dividend yields range from 1.07 % (PCG, which suspended its dividend during the bankruptcy) to 6.39 % (OKE), so the drift is generically non-zero and of the same order as the spread's own amplitude. **This is why the choice of price series is a modelling decision and not data preparation.**

3.6 What forward-fill does, and what it does not do

Section 7.2 reports that forward-fill does not manufacture mean reversion, contrary to the standard argument. Here is why, in two lines.

Proposition 4 (Forward-fill preserves the zero autocovariance of increments)

Let $X_t = X_{t-1} + \varepsilon_t$ be a random walk with i.i.d. centred increments, and let (G_t) taking values in $\{0,1\}$ be a gap indicator **independent of (ε_t)** . Define the forward-filled series $X_t^f = X_{\tau(t)}$ with $\tau(t) = \max\{s \leq t : G_s = 0\}$. Then for every t ,

$$\text{Cov}(\Delta X_t^f, \Delta X_{t+1}^f) = 0.$$

Proof.

Condition on the whole gap sequence G , which is independent of ε . Given G , the increment ΔX_t^f equals 0 if $G_t = 1$, and otherwise equals the sum of the ε_s over the deterministic index block $(\tau(t-1), \tau(t)]$. Three cases exhaust the possibilities. If $G_t = 1$ then $\Delta X_t^f = 0$ and the product vanishes; if $G_{t+1} = 1$ then $\Delta X_{t+1}^f = 0$ and it vanishes likewise; otherwise the two increments are sums of ε over **disjoint** index blocks, hence uncorrelated. In all cases the conditional covariance is 0, and the tower property gives the unconditional result. ■

The intuition the proof formalises: forward-fill **redistributes** increments across dates without creating or destroying any. The missing day carries a zero return and the next observed day carries the accumulated sum. Nothing compensatory happens, so no mean reversion appears. What is genuinely damaged is tradability — signals land on dates where the security did not trade — which is an execution issue, not a statistical one.

3.7 Survivorship as a conditioning bias

Section 8.1 asserts that survivorship bias is a form of look-ahead whose sign is known. The following makes both claims precise.

Proposition 5 (Sign of the survivorship bias)

Let R be the return of a strategy over the sample, and let S be the event that a name survives to the end of the sample (no bankruptcy, no acquisition). A universe built from current constituents estimates $E[R | S]$, whereas the quantity of interest is the unconditional $E[R]$. By the law of total expectation,

$$E[R] = E[R | S] P(S) + E[R | S^c] P(S^c),$$

so that

$$E[R | S] - E[R] = P(S^c)(E[R | S] - E[R | S^c]).$$

Hence the bias is positive if and only if $E[R | S] > E[R | S^c]$.

Proof.

The displayed identity is the law of total expectation rearranged; the sign claim follows because $P(S^c) \geq 0$. ■

The empirical content is the sign of $E[R | S] - E[R | S^c]$. Every one of the five names lost from our universe left the index by **acquisition**, and a takeover breaks a cointegration relation in the most damaging possible way: the target's price jumps to the offer and then freezes, so a short leg on the target takes an instantaneous loss that never reverts. Therefore $E[R | S^c]$ much smaller than $E[R | S]$, the bias is positive, and the naive universe **overstates** performance. A favourable bias is the dangerous kind: nothing in the results invites you to look for it.

3.8 How many false discoveries a screener produces

Section 10.1 quotes an expected number of spurious discoveries. That figure rests on two elementary facts worth stating, because the second is stronger than people expect.

Proposition 6 (Probability integral transform)

Let T be a test statistic with continuous distribution function F under H_0 , and let $p = 1 - F(T)$ be the associated p-value. Then p is uniform on $[0, 1]$ under H_0 , so $P(p \leq \alpha) = \alpha$ for every α in $[0, 1]$.

Proof.

For u in $[0, 1]$, $P(p \leq u) = P(1 - F(T) \leq u) = P(F(T) \geq 1 - u)$. Since F is continuous, $F(T)$ is uniform on $[0, 1]$, hence this equals $1 - (1 - u) = u$. ■

Proposition 7 (Expected false discoveries, under arbitrary dependence)

Test N null hypotheses, all true, at level α , and let $V = \text{sum over } i \text{ of } I\{p_i \leq \alpha\}$ be the number of rejections. Then

$$E[V] = \alpha N,$$

with no independence assumption whatsoever.

Proof.

By linearity of the expectation and Proposition 6,

$$E[V] = \sum_{i=1}^N P(p_i \leq \alpha) = \alpha N.$$

Linearity holds for arbitrarily dependent summands, which is the point. ■

The part that does need independence

Only the **expectation** of V is dependence-free. Its variance is not:

$\text{Var}(V) = N\alpha(1-\alpha) + \text{the sum over } i \neq j \text{ of } \text{Cov}(I\{p_i \leq \alpha\}, I\{p_j \leq \alpha\})$, and the covariance term dominates when the tests share structure.

In our screener the pairs overlap by construction — every pair containing a given name is correlated with every other pair containing it — so V is far more dispersed than a binomial count. Phase 2 measures the consequence: one name holds 16 % to 50 % of the selected pairs in every fold. This is exactly why the Bonferroni-versus-Benjamini-Hochberg choice of Phase 3 is substantive rather than cosmetic: the former is valid under arbitrary dependence, the latter is not.

With $N = 465$ pairs and $\alpha = 5\%$, Proposition 7 gives 23 expected spurious discoveries. The screener finds 47. Roughly half of what looks like a discovery is, in expectation, noise — and that is the observation Phase 3 is built to address.

4. Why Phase 1 deserves a whole document

Pairs trading has a poor reputation among quantitative funds, for good reason: it is the exercise where producing a spectacular and entirely false backtest is easiest. Three mechanisms suffice, and all three play out in Phase 1 or just after.

1. **Survivorship bias.** You build the universe from companies that exist today, so you only ever trade survivors.
2. **Multiple testing.** You test hundreds of pairs and keep the best, without correcting for the fact that if you test enough, you always find something.
3. **Execution or normalisation look-ahead.** You use, to decide at date t , a statistic computed on data after t .

Point 2 is handled in Phase 3, point 3 in Phases 2 and 6. **Point 1 is settled entirely in Phase 1**, and it is the only one that cannot be fixed afterwards: if the universe is wrong, everything downstream is contaminated.

What makes these mistakes different

A normal software bug announces itself: the program stops, an exception is raised, a test fails. Phase 1 mistakes do none of that.

They produce a backtest that runs normally, smooth curves and a presentable Sharpe ratio. The only way to catch them is to look for them deliberately, by writing checks whose sole purpose is to fail.

5. Decision 1 — The universe: S&P 500 utilities

5.1 First sub-decision: stay inside a single sector

We could have formed pairs across the whole S&P 500. We chose to restrict to a single GICS sector. Two justifications, of different kinds.

Economic justification

Section 2.3 established that cointegration requires a shared stochastic trend. Within a sector you can name it: for utilities, it is exposure to long rates (these are very high-duration assets, bought for yield), the regulatory regime (tariffs set by commissions on an asset base), and input costs (gas, fuels). Across sectors that mechanism does not exist — cointegration between a bank and a semiconductor maker would be a sample artefact.

Statistical justification

The number of pairs grows as N^2 . Across 500 names that is $500 \times 499 / 2 = 124,750$ tests to correct for. Under a Bonferroni correction the threshold becomes $\alpha/124750 \approx 4 \times 10^{-7}$: no pair would ever pass, and Phase 3 would have nothing to show. Restricting the sector brings the problem to a scale where correction is **binding but not destructive** — 465 tests.

This matters for interviews: the sector restriction is not a shortcut to reduce workload. It is a **reduction of the search space grounded in a prior**, which is precisely what a serious quantitative researcher does before running a screener.

5.2 Second sub-decision: which of the eleven sectors

Four criteria were fixed **before** looking at any data, which is the decisive element: picking the sector after seeing which one gives the best P&L would be multiple testing at the sector level, and would destroy the project's own argument.

Criterion	What it requires	Why
Identifiable common factor	Being able to name the economic mechanism linking the names	Without it, any cointegration found is a sample artefact
Sufficient size	At least ~25 names, i.e. ≥ 300 pairs	Below that, the multiple-testing demonstration carries no weight
Datable, explainable break	At least one episode where the relation breaks, with a known cause	V1 must explain <i>why</i> it breaks, not merely observe that it did
Execution homogeneity	One market, one currency, one closing time	Avoids illusory predictability from asynchronous closes

The four sector-selection criteria, fixed before any examination of the data.

Utilities satisfy all four. The common factor is the clearest in the entire US equity market. The sector holds between 25 and 31 names depending on the date. And it offers not one but **three tellable breaks**:

- **March 2020, COVID.** A generalised liquidity shock: every correlation converges to 1 and long-run relations are temporarily drowned out.
- **2022, rate shock.** The rapid rise in rates hits utilities through their duration, but **unevenly**, depending on leverage and debt structure. The dispersion of sensitivities breaks relations that had held for years.
- **2023-2024, AI repricing.** Data-centre electricity demand abruptly revalues *merchant* generators — those selling at market prices (VST, CEG, NRG) — while regulated names (ED, WEC) stay capped by their tariffs. This break is **structural**: the two groups permanently stop sharing a common trend.

The 2023-2024 break is the narrative core of V1

It has everything an interviewer wants: a date, a precise economic mechanism, a verifiable statistical consequence (β drifts, half-life explodes, ADF p-value rises), and a **permanent** character — this is not a temporary shock to wait out, it is a relation that no longer exists and must be exited.

5.3 What was ruled out, and why

Sector	What it offered	Why ruled out
Regional banks (KRE)	~90-140 names, so ~4,000 pairs: far more spectacular multiple testing. And March 2023 (SVB, Signature, First Republic) makes survivorship bias the central subject rather than a footnote.	Prices for bankrupt companies are not obtainable for free. The budget would have gone into data plumbing instead of methodology.

Sector	What it offered	Why ruled out
Energy E&P (XOP)	Obvious common factor (the oil price) and a spectacular 2020 break, with WTI going negative in April.	The 2023-2024 M&A wave truncates many series. Heterogeneous leverage makes cointegration fragile.
Semiconductors	The 2023 AI break is extremely visible.	Too trending: a real risk of finding almost no cointegrated pair at all, which would empty Phase 3 of content.

5.4 The objection you will get

"Your spread is just a disguised duration bet"

The objection is serious and partly right: if every name reacts to rates, a long/short between two of them still carries the differential rate sensitivity.

The right answer is not to deny it, it is to measure it. The hedge ratio absorbs the *common* exposure by construction. What must be checked is that the **residual** — i.e. the P&L — is no longer explained by rates. The test is explicit: regress daily strategy returns on changes in the US 10-year yield and show the coefficient is not significant. That is scheduled for Phase 7.

Answering "I don't know, but here is the test I would run" is worth far more than denying the objection.

6. Decision 2 — Sample period and time split

6.1 Why 2014 → 2026

The period results from a trade-off between two opposing forces.

Lengthening history increases the number of independent sub-periods, hence the reliability of conclusions: with three years of data you cannot say anything about a strategy's stability. **Shortening** history improves data quality (survivorship worsens as you go back) and relevance (the sector's structure in 2010 — before the renewables boom, before the Constellation spin-off — bears little resemblance to today's).

Twelve and a half years puts the cursor where the two effects balance: enough sub-periods to discuss signal decay, not so many that you test a strategy on a sector that no longer exists.

6.2 Walk-forward: three years of selection, one year of trading

This is the protocol guaranteeing that pair selection and pair trading never happen on the same data.

```
fold 1 :  SELECTION 2014-2016 (3 yrs)  ->  TRADING 2017 (1 yr)
fold 2 :  SELECTION 2015-2017          ->  TRADING 2018
fold 3 :  SELECTION 2016-2018          ->  TRADING 2019
...
fold 9 :  SELECTION 2022-2024          ->  TRADING 2025
```

```
Each fold: point-in-time universe recomputed, liquidity filter
recomputed, cointegration retested, pairs selected -- then FROZEN.
The trading window NEVER influences the selection preceding it.
```

Why three years of selection? A cointegration test needs enough points to separate a genuine equilibrium relation from chance: roughly 750 trading days. Over one year the augmented Dickey-Fuller test has almost no power — it rarely rejects non-stationarity even when it is false.

Why one year of trading? Beyond that you implicitly assume a relation selected in 2016 still holds in 2020. That is exactly the assumption V1 disputes. One year balances re-selection frequency against turnover cost.

The trap not to fall into

Once in the trading window it is tempting to retest cointegration and keep only the pairs that "still work". That is a **direct reintroduction of look-ahead**: you would use information from the trading period to decide what to trade during that same period.

The rule is absolute: **selection is frozen at the end of the in-sample window**. If a pair degrades during trading, it is the regime-detection mechanism (V1) that must exit it — and that mechanism, too, looks only at the past.

7. Decision 3 — Which prices, exactly

7.1 Raw, split-adjusted, or total-return?

Three different price series exist for the same stock, and the choice among them is not cosmetic.

- The **raw price** is what shows on screen. It drops mechanically on a split or an ex-dividend date.
- The **split-adjusted price** corrects share splits.
- The **total-return price** corrects splits *and* dividends: it represents the value of an investment that reinvests its dividends.

We chose total-return. Here is the argument, because it is the kind of point that immediately separates a serious candidate.

Why it is decisive specifically for utilities

Utilities pay 3 to 4 % dividend yield per year, and crucially that yield is **widely dispersed** across names. Write δ_A and δ_B for the annual dividend yields of two names. The raw price separates from the total-return price by a factor growing with time:

$$\log P^{raw}(t) \approx \log P^{TR}(t) - \delta \cdot t$$

A spread computed on raw prices therefore carries a term $-(\delta_A - \delta_B) \cdot t$: a **deterministic linear drift**. A series that drifts linearly is not stationary. The ADF test will reject it — and we would discard **genuinely cointegrated** pairs merely because their dividend policies differ.

Magnitude measured on the actual universe: implied dividend yield ranges from **1.07 % (PCG) to 6.39 % (OKE)**, a 5.3-point spread between two names in the same sector. A 2-point gap already injects about 6 % of drift over a three-year selection window — comparable to the amplitude of the spread itself. The effect is not marginal.

The experiment, and what it actually licenses us to conclude

We tested all 465 pairs on a representative in-sample window (2016-2018), once on total-return prices and once on raw prices, and compared the resulting **selections**:

```
465 pairs tested on 2016-2018 (Engle-Granger, 5 % threshold)
```

```
significant on TOTAL-RETURN : 112
```

```
significant on RAW PRICES : 137
```

```
disagreement : 51 pairs (11 %) -- 13 lost, 38 created
```

```
but most of it is p-value noise near the threshold:
```

```
lost : 11/13 marginal (85 %)
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```
created : 26/38 marginal (68 %)
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```
DECISIVE FLIPS (p > 0.15 one side, < 0.05 the other)
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```
pairs MANUFACTURED by raw prices : 16
```

```
pairs DESTROYED by raw prices : 1
```

This result must be read carefully, and that is an important methodological point. The tempting conclusion — "raw prices manufacture 38 false pairs" — is not defensible: 68 % of those pairs have a p-value in [0.03, 0.08]. They flip because a p-value near the threshold is noisy, not because dividend drift changed anything substantial. Counting them as selection errors would be dishonest.

What survives is the asymmetry of decisive flips: 16 against 1. Sixteen pairs with no equilibrium relation on the economically correct prices become decisively significant once raw prices are used; only one flips the other way. The correlation between drift magnitude and p-value gap is 0.29 — the mechanism is there, but it explains only part of the phenomenon.

The mechanism, for the decisive cases. If two names diverge economically but the one rising faster also pays the larger dividend, the drift **cancels** the divergence on raw prices. The spread becomes artificially flat, hence "stationary" as far as the test is concerned.

The formulation to keep

The two price series produce selections differing on 11 % of pairs. Saying "38 false pairs" would overstate it — most is threshold noise. But the **16 against 1** asymmetry is robust: raw prices manufacture apparent cointegration far more often than they destroy it.

The choice of price series is not a data-preparation detail: **it is a decision that changes the selection itself.**

Being able to separate "what I would like to conclude" from "what my data licenses" is precisely what a fund probes for in interviews.

Total-return is also the right **economic** object: a long/short position genuinely collects the dividend on the long leg and pays it on the short leg. The simulated P&L must reflect that.

The residual leak you must be able to name

Adjusted prices are **retro-adjusted**. Every new dividend paid retroactively modifies the entire historical series. Put differently: the 2018 series as I download it in 2026 is not the one a trader observed in 2018.

That is formally a leak: the series embeds later information. Its effect is second-order here, since both legs are adjusted in the same direction and the position is long/short. But it exists, and **naming it spontaneously** is exactly what separates someone who has thought about it from someone who followed a tutorial.

7.2 Why missing prices are never filled

When a series has a gap — partial holiday, trading halt, vendor failure — the universal reflex is *forward-fill*: carry the last known price. We do not. But the reason is not the one you read everywhere, and this section deserves close attention because it contains **a mistake made and then corrected**.

The usual argument — and why it is wrong

The argument found in most tutorials, and which I advanced myself before testing it, runs: carrying a price forward creates a zero return followed by a catch-up, hence the sequence "no move, then compensating move", which is the signature of mean reversion. Forward-fill would therefore manufacture the very signal the strategy looks for.

That is false, and simulation shows it unambiguously. Simulate a pure random walk — hence no mean reversion whatsoever — introduce 5 % missing days, forward-fill, and test:

```
Pure random walk, 400 simulations, 5 % missing days

ADF rejection rate at 5 %   without ffill :    6.8 %
ADF rejection rate at 5 %   with ffill    :    6.5 %
lag-1 autocorr. of diffs    without ffill : -0.0016
lag-1 autocorr. of diffs    with ffill    : -0.0017
```

No effect. And on a genuinely mean-reverting spread (Ornstein-Uhlenbeck with a 15-day half-life), forward-fill biases neither the estimated half-life (+0.2 %) nor the number of signals generated (67.6 versus 67.8).

The reason is simple once seen: forward-fill **redistributes** the increments without modifying them. The missing day carries a zero return, the next day carries the sum of both increments. The total is unchanged and, since increments are independent, the covariance between consecutive returns stays zero. There is no "compensating catch-up": there is a deferral.

What this means for interviews

If you advance the "manufactured mean reversion" argument to someone who has run the simulation, you lose all credibility on the rest of the project.

The correct argument is simpler and more solid — and being able to say "I believed it, I tested it, it was false, here is what is true" is worth more than any memorised justification.

The correct argument: tradability

The real problem with forward-fill is not statistical, it is **operational**. On an Ornstein-Uhlenbeck spread with 5 % missing days:

```
OU spread (15-day half-life), 5 % missing days, 300 simulations

estimated half-life  without ffill : 13.95 d
estimated half-life  with ffill    : 13.98 d      (bias +0.2 %)
signals |z| > 2       without ffill : 67.8
signals |z| > 2       with ffill    : 67.6

signals landing on a NON-TRADING day : 2.6 per series
i.e. 3.8 % of signals, at a price that never existed
```

The justification to keep

Forward-fill does not corrupt the spread's statistics: it corrupts **execution**. About 4 % of entry signals land on a day the name did not trade. The backtest then opens a position at a price that never existed, and books P&L on an impossible trade.

This is an execution leak, of the same family as trading at the same day's close on which the signal was computed — the subject of Phase 6.

Decision: no forward-fill. Missing values stay NaN and are handled explicitly, pair by pair, at test time.

7.3 Cleaning, and why it is deliberately minimal

Cleaning reduces to two rules, chosen to be **non-destructive** — aggressive cleaning is itself a source of bias.

1. **Drop entirely empty columns.** These are tickers the vendor no longer serves, typically delisted companies. They are not erased from the project's memory: they are accounted for in the survivorship report (section 10).
2. **Drop dates where fewer than 80 % of live names trade.** These are partial holidays or data incidents, not sessions. The subtle point is the denominator: we compare against the number of names **alive on that date**, not the total column count. Otherwise a name listed in 2022 would fail the criterion on every session from 2014 to 2021 and we would delete eight years of valid data.

8. Decision 4 — The point-in-time universe

8.1 The problem: why survivorship is a leak

Taking an index's **current** constituents and pulling their history is the method used in the vast majority of student projects. It is wrong, and the reason runs deeper than "a bias".

Selecting today the companies in the S&P 500 means using information dated today — **the fact that these companies still exist and are large enough to be in the index** — to decide what to trade in 2015. That is structurally identical to look-ahead on the signal. The only difference is that the leaking variable is not a future price: it is **survival**.

The effect is systematic and always in the same direction: you exclude exactly those companies that went bankrupt, were acquired, or fell apart — the cases where a cointegration relation breaks violently. You measure performance on a sample stripped of its worst scenarios.

8.2 Two possible methods, and why we switched

Method A — reconstruct from changes. Start from today's list and walk back in time inverting each addition and removal recorded in a historical table. That was the original plan.

That table no longer exists on the public source, so we switched to a **Method B, which turns out to be strictly superior**: read the page's revision history. The page listing S&P 500 constituents has been continuously edited for more than fifteen years, and every past version remains accessible. We therefore retrieve the page **as it existed** on each selection date.

	Method A — change table	Method B — revisions (chosen)
Principle	Invert changes backwards from today	Read the page as it stood on the target date
GICS sector	Lost for exited companies: the table records only the ticker, the sector would have to be reassigned by hand	Directly observed , as classified at the time

	Method A — change table	Method B — revisions (chosen)
Reclassifications	Invisible: a name moved between sectors is not an index change	Captured automatically (the ONEOK case, section 9.4)
Robustness	One error in the table propagates to every earlier date	Each date is read independently: an error stays local

Switching to Method B was not a fallback: it yields membership *and* point-in-time sector, both directly observed.

8.3 The technical detail that guarantees no leak

The entire no-leak property rests on one request parameter:

```
params = {
    "rvstart": timestamp,    # the selection date
    "rvdir":    "older",     # <-- walk BACK in time from that date
    "rvlimit": 1,
}
```

`rvdir="older"` asks for the last revision published **before** the selection date. With `"newer"` you would get the first revision published *after* — a page that may already reflect index changes later than the decision. **One word separates a correct reconstruction from a leak.**

The observed gap between selection date and the revision used is **2 days median, 19 days maximum**. That gap always runs conservative: the page used is slightly *behind* reality, never ahead.

8.4 Three implementation sub-decisions

Semi-annual cadence

We snapshot the index every six months. Since walk-forward re-selection is annual, each selection date has a snapshot at most six months old. Going monthly would multiply requests sixfold for precision the strategy cannot use: it rebalances once a year.

Systematic disk cache

Every downloaded revision is written to disk. This is not an optimisation, it is a **reproducibility requirement**: without a cache, universe construction depends on a web page editable at any moment, and even retroactively. Two runs a month apart could yield two different universes, hence two different backtests, with no code change whatsoever.

Failing loudly on a failed request

If a request fails repeatedly the program raises and stops. The temptation would be to skip the missing snapshot and continue. That would be exactly the kind of silent degradation the whole of Phase 1 is built against: a universe truncated over one period, with nothing flagging it.

9. Decision 5 — Ticker identity checking

9.1 What we found

Once the point-in-time universe was built, we downloaded prices for the 38 tickers. Inspecting the start dates revealed a problem we had not anticipated and which would have invalidated the entire project.

Ticker	Company in the index	Exit	Data served from	What it actually is
NU	Northeast Utilities	2015	9 December 2021	Nu Holdings — Brazilian neobank, IPO late 2021
POM	Pepco Holdings	2016 (acquired by Exelon)	8 October 2025	An unrelated company, listed since 2025
TE	TECO Energy	2016 (acquired by Emera)	10 January 2020	An unrelated company
TEG	Integrus Energy	2015 (acquired by WEC)	22 December 2015	An unrelated company, ticker reused months later

The four ticker-recycling cases detected in the universe.

Tickers are not durable identifiers. When a company disappears its symbol returns to the pool and can be reassigned. Three letters do not identify a company — they identify a slot on a market at a point in time.

What would have happened without this check

The backtest would have treated the prices of a **Brazilian neobank** as those of a regulated US utility. It would have tested cointegration between Nu Holdings and Duke Energy, possibly found it significant over some window, and opened positions.

No exception raised. No test failed. The P&L curve perfectly smooth. And in the interview you would have confidently defended an entirely false result.

This is the exact illustration of section 4's thesis: **the mistakes that matter do not raise errors.**

9.2 The detection rule

The idea: we hold two independent pieces of information about each ticker. On one side its **membership window**, from the point-in-time reconstruction. On the other its **quotation period**, from the price vendor. If the ticker denotes the same company in both sources, the two intervals must overlap.

$$\text{overlap} = |[first\ quote,\ last\ quote] \cap [index\ entry,\ index\ exit]| / |[index\ entry,\ index\ exit]|$$

- **near-zero** overlap → the ticker denotes another company: `RECYCLED`
- **partial** overlap (< 50 %) → truncated history, manual review: `SUSPECT`
- **substantial** overlap → identity consistent: `OK`

The check is **blocking**: tickers flagged `RECYCLED`, `SUSPECT` or `NO_DATA` are removed from the tradable universe, and the exclusion list is written to an auditable report.

9.3 The two bugs found while writing this check

They are worth telling, because they show that even an anti-error check must itself be verified.

Bug 1 — the ONEOK false positive

The first version flagged `OK` as recycled. But `ONEOK` is the same company, with a complete history. The error came from the membership-window definition: `[first member snapshot, last member snapshot]`. `ONEOK` appears in only **one** snapshot — GICS reclassified it from *Utilities* to *Energy* in early 2014 — so its window had zero length and the overlap divided by zero.

Fix: being a member at date t means being one at least until the next snapshot, so the window is widened by six months on the right. The `ONEOK` case is interesting in itself: it is a **sector reclassification**, which Method B captures automatically while Method A would have missed it entirely.

Bug 2 — the exact-zero threshold

The initial rule flagged as recycled an overlap of **exactly** zero. `TEG` has an overlap of 0.014: Integrys was acquired in June 2015 and the ticker reassigned by December, so the two intervals overlap by ten days. With a zero threshold, `TEG` slipped through.

Fix: threshold at 5 %. The general lesson is that a binary test resting on exact equality is fragile — you need a margin, and you need to have looked at the data to calibrate it.

9.4 The blind spot that remains

The check compares two intervals. It would therefore poorly detect a recycling occurring **during** the membership window — the overlap would stay high and the status would read `OK`. That case is not observed here, and is unlikely in practice (a ticker is not reassigned instantly), but it is documented rather than passed over.

A genuinely robust check would use a durable identifier — CUSIP, SEDOL, or CRSP's PERMNO — which does not change when the ticker does. That is exactly what professional databases provide, and one reason they are expensive.

10. Phase 1 results

10.1 The numbers

Quantity	Value	What it is for
Tickers in raw union	38	Every name that belonged to the sector over the period
Tradable tickers	33	After the identity check
Sector size by date	25 to 31	Varies with index entries and exits
Maximum testable pairs	465	$N(N-1)/2$ — Phase 3's multiple-testing load
Expected false positives at $\alpha = 5\%$	$\approx$ 23	Phase 3's opening figure
Trading sessions	3140	02/01/2014 → 29/06/2026
Index snapshots	25	Semi-annual, 2-day median lag

The figure to remember for Phase 3

With 31 names you test 465 pairs. Under the null — no pair is truly cointegrated — a 5 % test would still declare $\approx$ **23 significant**, by pure chance.

Phase 3's entire argument consists of comparing that number to the pairs actually found significant. If you find 30, you have found almost nothing.

10.2 The PG&E case: what point-in-time actually buys

The reconstruction shows PCG is **out of the index from July 2019 to July 2022**: PG&E filed for Chapter 11 in January 2019 after the California wildfires, was removed from the S&P 500, then readmitted after emerging.

Since PCG is a member **today**, a universe built on current constituents would include it throughout, and would therefore trade it **during its bankruptcy** — implicitly knowing it came out alive and its price eventually recovered. That is look-ahead in its purest form, on one of the sector's most violent episodes.

Point-in-time reconstruction excludes it automatically, with no rule written for this case. That is the sign we fixed the cause and not the symptom.

10.3 Residual survivorship, quantified and signed

Of 38 tickers in the union, **5 are excluded**: GAS, NU, POM, TE, TEG.

Ticker	Status	What actually happened
GAS	No data	AGL Resources, acquired by Southern Company (2016). The vendor serves nothing: outright loss.
NU	Recycled	Northeast Utilities, renamed Eversource (ES) in 2015. Only an apparent loss: the ES series goes back to 2014 and carries the full history.
POM	Recycled	Pepco Holdings, acquired by Exelon (2016).
TE	Recycled	TECO Energy, acquired by Emera (2016).
TEG	Recycled	Integrus Energy, acquired by WEC (2015).

By volume: 732 membership slots (snapshot × ticker) in the raw universe, 708 retained, i.e. **3.3 % lost**. Losses are **entirely concentrated in 2014-2016** and **zero from January 2017 onward**. At the worst snapshot, 5 names are missing out of 31, roughly 30 % of potential pairs untestable.

The direction of the bias — the most important point in this section

All five lost names left the index **by acquisition**, none by bankruptcy. And a takeover is precisely the event that breaks a cointegration relation most brutally: the target's price jumps to the offer price, then freezes, permanently decorrelated from its sector. A short position on the target at announcement takes an instant loss that **never** reverts.

The formulation to keep

Excluding these five names removes extreme, irreversible loss scenarios from the sample. **The residual bias therefore overstates the strategy's performance.**

This is the most dangerous case: a favourable bias never announces itself. You have no reason to look for it, because the results look good.

Practical consequence: the first fold (selection 2014-2016) is the most exposed. Honesty means reporting its result **separately** rather than silently aggregating it with the others.

10.4 Two grey areas, acknowledged

- **EVRG.** Evergy was formed in 2018 from the Westar / Great Plains merger. The vendor serves history back to 2014: those are the predecessor's prices. Defensible — the predecessor is the relevant economic object — but worth knowing if an interviewer digs.
- **SCG.** SCANA shows an overlap of 0.69: served history starts in July 2015 while the company was a member from 2014. History truncated, identity consistent. Kept, with less data than the others.

11. What was deliberately deferred

11.1 The liquidity filter

The plan calls for a universe of **liquid** stocks. Yet no liquidity filter was applied in Phase 1. That is deliberate, and the reason is a leak.

Filtering now would mean computing median volume over the **whole** 2014-2026 period and keeping names above a threshold. You would then be using, to decide what to trade in 2015, the information "this name will still be liquid in 2025". That information is strongly correlated with the firm's survival and success: it is survivorship bias dressed up as a technical criterion.

The filter must be recomputed at each fold, on the in-sample window only, exactly like pair selection. It therefore arrives in Phase 3, where selection itself is built. That is why Phase 1 ends on an unfiltered universe — it is not an oversight.

11.2 Guardrails currently in place

Guardrail	Against what	Where
Point-in-time membership	Survivorship in the composition	src/universe.py
Blocking identity check	Ticker recycling	validate_identity
Total-return prices	Dividend drift in the spread	src/data.py
No forward-fill	Signals on non-trading days	src/data.py
"Live names" denominator	Deleting valid sessions	_clean
Disk cache	Non-reproducible universe	data/raw/wiki/
Loud failure on failed request	Silently truncated universe	_get
Dated universe decision	Sector-level data snooping	docs/01_universe_decision.md
Deferred liquidity filter	Survivorship disguised as a technical criterion	Phase 3

12. The two questions — detailed answers

Question 1 — The identity check's blind spot

"Which recycling case would slip through the rule? And by what concrete mechanism would the P&L become wrong without any test catching it?"

The case that slips through

A recycling occurring **during** the membership window. The rule measures overlap between quotation period and membership window; if company X is a member from 2014 to 2020 and disappears in 2017, a ticker reassigned in 2018 to a company Y would produce a continuous series from 2014 to 2020, perfectly covering the window. Status: OK.

How the P&L gets contaminated

The concatenated series would show a **level discontinuity** at the switch date: the price jumps from X's to Y's, with no economic link. Three cascading consequences:

1. The in-sample cointegration test would see a spread containing a jump. Depending on where the jump sits, the test would **reject** wrongly (the jump resembles a unit root) or **accept** wrongly if the jump is followed by a stable regime resembling mean reversion.
2. If the pair gets selected, the z-score at the jump would reach an extreme value — typically $|z| > 5$. The strategy would open a **maximum** position on what it reads as an exceptional opportunity.
3. That position never unwinds, because the jump is not a temporary deviation but a change of object. The P&L books a massive, purely fictitious gain or loss.

Why no test catches it: every Phase 7 diagnostic (Sharpe, drawdown, hit rate, turnover) is an *aggregate* statistic computed **on that same corrupted series**. They are mutually consistent and consistent with the data. Nothing is incoherent — the data is simply wrong. No amount of internal checking can detect an error that affects the system's input.

What would catch it: a durable identifier (CUSIP, SEDOL, PERMNO), or failing that a **return-discontinuity check** — a daily return of several hundred percent not matched by a known split is the signature of an entity change. That is a cheap check and would be reasonable to add.

Question 2 — The PG&E objection

"A real fund is not constrained by S&P 500 membership; it could have traded PG&E during the bankruptcy. By excluding it you have not corrected a bias, you have introduced another."

The objection is right on the facts and wrong in its conclusion. Concede the first point before answering.

What index membership actually stands in for

We do not use S&P 500 membership because a mandate requires it. We use it as an **observable, point-in-time proxy** for a set of properties we genuinely want to impose and which are hard to measure directly: sufficient market cap, sufficient free float, liquidity allowing entry and exit without impact, and the practical ability to **sell short** — that is, the existence of a borrowable inventory at a reasonable cost.

That last point is decisive. Pairs trading requires a short leg. During a bankruptcy proceeding a stock becomes extremely hard and expensive to borrow: borrow costs can reach tens of percent annualised, and the lender can recall at any time, forcing a buy-in at the worst moment (a short squeeze). A backtest that ignores those costs and prices them at zero **massively overstates** performance.

The answer in three steps

1. **Concede.** Yes, index membership is not a fund's real constraint. It is an approximation.
2. **Turn the argument around.** It is a **conservative** approximation. It excludes precisely the periods where execution — especially stock borrow — is costliest and least certain. Including PG&E during its bankruptcy with normal assumed financing costs would not be more realistic: it would be **more optimistic**, pricing the market's most expensive risk at zero.
3. **Name the real fix.** The rigorous solution is not to include PG&E carelessly, but to model borrow cost and availability explicitly. That requires securities-lending data, which is paid. Absent it, exclusion is the conservative choice, and I document it.

The general principle behind this answer

Faced with a well-founded methodological objection, the bad answer is to deny; the mediocre answer is to concede; **the good answer is to show which way the choice biases the result, and to systematically prefer the direction unfavourable to the strategy.**

That is the guiding line of all of Phase 1, and the message the project as a whole must convey.

13. Glossary

Term	Definition
Spread	Gap between two log-prices weighted by the hedge ratio: $\log P_A - \beta \log P_B$.
Hedge ratio (β)	Weight of the second leg. Answers: how much of B to sell per dollar of A.
I(0) / I(1)	Stationary series / series whose first difference alone is stationary. A stock price is typically I(1).
Stationarity	Property of a series whose law does not depend on time: it oscillates around a fixed mean rather than drifting.
Cointegration	Existence of an I(0) linear combination of I(1) series. Formalises a long-run equilibrium relation.
ADF	Augmented Dickey-Fuller test. Tests for a unit root; applied to the residual to test its stationarity.
Look-ahead	Using, to decide at date t , information unavailable at t . Inflates performance without announcing itself.
Survivorship bias	A form of look-ahead on existence: keeping only companies that survived to today.
Point-in-time	Data reconstructed as it was observable at the date considered, not as we know it today.
Walk-forward	Validation protocol where selection and evaluation follow each other in time, never overlapping.
In-sample / Out-of-sample	Selection and estimation period / test period, later and strictly disjoint.
Total-return	Price series adjusted for splits and reinvested dividends.
Forward-fill	Filling a missing value with the last known value. Rejected here (section 7.2).
Merchant generator	Power producer selling at market prices, as opposed to a regulated utility with set tariffs.
GICS	Standard sector taxonomy. Utilities is the level-1 sector used here.
Ticker recycling	Reassignment of a market symbol to another company after the first one disappears.